BEHAVIOR-BASED PHISHING ATTACK DETECTION USING ARTIFICIAL INTELLIGENCE

A Seminar Report

Submitted by

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ABSTRACT

Phishing attacks are among the most prevalent cyber security threats, targeting users by impersonating legitimate entities to steal sensitive information such as login credentials, financial details, and personal data. Traditional phishing detection techniques primarily rely on blacklist-based filtering and rule-based analysis, which are ineffective against rapidly evolving and newly generated phishing attacks. This report presents a behavior-based phishing detection approach using Artificial Intelligence (AI) to enhance detection accuracy and adaptability. The proposed system analyzes behavioral and structural characteristics such as URL patterns, redirection behavior, login form attributes, domain metadata, and webpage inconsistencies. Machine learning models are trained on labeled datasets to classify websites as phishing or legitimate. The approach improves real-time detection capabilities and supports identification of zero-day attacks. The report discusses system architecture, working mechanism, technology stack, applications, benefits, limitations, and future scope of AI-based phishing detection systems.

1. INTRODUCTION

The rapid growth of internet services and digital platforms has significantly increased cyber security risks. Among various cyber threats, phishing remains one of the most widespread and dangerous social engineering attacks. Phishing attacks involve impersonating trusted entities such as banks, social media platforms, and service providers to deceive users into revealing confidential information. These attacks are commonly delivered through fraudulent emails, fake websites, and malicious links. The consequences include financial loss, identity theft, and data breaches. Conventional security solutions struggle to address modern phishing techniques due to their dynamic nature. Artificial Intelligence-based detection methods offer adaptive learning capabilities that can effectively identify suspicious behavioral patterns.

1.1 Background and Motivation

Traditional phishing detection mechanisms rely heavily on blacklists and heuristic rules. However, attackers frequently generate new domains and modify attack strategies, rendering static defenses ineffective.
AI-driven systems provide automated pattern recognition and can detect anomalies across large datasets, making them suitable for combating evolving phishing threats.

1.2 Phishing Detection Systems

Phishing detection systems are designed to identify malicious websites and deceptive communication channels. Early systems focused on URL filtering and signature matching. Modern approaches incorporate machine learning and behavioral analysis to improve detection performance and adaptability.

2. PROPOSED SYSTEM

2.1 Problem Definition and Requirements

The dynamic nature of phishing attacks presents significant challenges for traditional detection systems. Static methods fail to identify newly generated phishing websites that do not match existing signatures. Therefore, an adaptive detection mechanism that analyzes behavioral patterns is required to enhance detection accuracy and scalability.

2.2 Overview of the Proposed System

The proposed system employs a behavior-based phishing detection approach using Artificial Intelligence. Instead of relying solely on static indicators, the system evaluates dynamic behavioral attributes extracted from websites and user interactions. Machine learning algorithms are trained to classify websites based on learned patterns.

2.3 System Architecture

The system architecture includes data collection modules, feature extraction engines, machine learning classifiers, and decision layers. Website datasets containing phishing and legitimate samples are processed to extract meaningful features for model training and evaluation.

2.4 Mechanism of the Proposed System

The system operates through a structured pipeline involving data collection, preprocessing, feature extraction, model training, classification, and alert generation. Trained models analyze incoming data and generate warnings for suspicious websites in real time.

2.5 Technology Stack

The system can be implemented using Python for machine learning development, along with libraries such as Scikit-learn, TensorFlow, or PyTorch. Web frameworks may be used for deployment and visualization of results.

2.6 Applications

Applications include browser-based phishing detection, email filtering systems, banking security platforms, enterprise cyber security tools, and online transaction monitoring systems.

2.7 Benefits and Drawbacks

The approach offers improved detection accuracy and adaptability to evolving threats. However, challenges such as false positives, computational overhead, and dataset quality must be addressed.

2.8 End Users and Deployment

The system can be deployed as a standalone application, browser extension, or cloud-based security service. End users include individuals, enterprises, and financial institutions requiring enhanced phishing protection.

3. CONCLUSION AND FUTURE SCOPE

3.1 Conclusion

Behavior-based phishing detection using Artificial Intelligence provides an effective approach for addressing modern phishing threats. By leveraging machine learning and behavioral analysis, the system enhances detection accuracy and supports proactive cyber defense mechanisms.

3.2 Future Scope

Future enhancements may include integration with deep learning architectures, real-time browser extensions, federated learning for privacy-preserving detection, and explainable AI techniques for improved transparency.

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